

CASCADE FAILURE

Analysis Report

Region: roorkee_upper_ganga | Timestep: 14 | Generated: 2026-05-20 11:39:30

Overall Severity: CRITICAL | Cascade Nodes Fired: 4 / 4

EXECUTIVE SUMMARY

A CRITICAL severity cascade failure sequence was detected in region roorkee_upper_ganga at simulation timestep 14. The cascade propagated through 4 node(s): SlopeStability → RiverBlockage → DamBreach → FloodWave.

Flood wave will reach village_A in approximately 2.4 minutes. Immediate evacuation of downstream settlements is recommended.

SIMULATION PARAMETERS

Parameter	Value	Unit
Soil Saturation	92.0	%
Slope Angle	38.0	degrees
Slope Area	20,000	m2
Rainfall	250.0	mm
Channel Width	20.0	m
Channel Depth	4.0	m
Upstream Flow	150.0	m3/s
Lake Rise Rate	1.20	m/hr
Manning's n	0.0350	-

Channel Slope	0.0030	m/m
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CASCADE CHAIN

The following nodes fired in sequence. Each node's output becomes input to the next node in the chain.

Node 01 SlopeStability — FIRED

Soil saturation exceeded failure threshold causing debris mobilization

Message: Slope failure - saturation 92.0%, slope 38.0 deg. Debris 20160 m3 moving downhill.

Severity: CRITICAL | Probability: 77.1%

Soil Saturation: 0.920
Slope Angle Deg: 38.000
Debris Volume M3: 20160.000
Failure Depth M: 3.360
Rainfall Mm: 250.000



Node 02 RiverBlockage — FIRED

Debris volume sufficient to block river channel forming upstream lake

Message: River channel 100% blocked. Lake forming upstream - breach risk in 0.0h.

Severity: CRITICAL | Probability: 100.0%

Blockage Ratio: 1.000
Debris Volume M3: 20160.000
Channel Width M: 20.000
Lake Formation Hours: 0.000
Upstream Flow M3S: 150.000



Node 03 DamBreach — FIRED

Debris dam structural failure releasing stored lake volume

Message: Debris dam breaching - lake rising 1.20 m/hr. Peak 1657 m3/s in ~42 min.

Severity: CRITICAL | Probability: 60.0%

Lake Rise Rate M Per Hr: 1.200
Peak Discharge M3S: 1657.000
Breach Width M: 33.600
Time To Peak Min: 42.000
Lake Volume M3: 80000.000
Dam Height M: 12.000



Node 04 FloodWave — FIRED

High-velocity flood wave propagating downstream to settlements

Message: Flood wave downstream - peak 1657 m3/s, speed 34.5 m/s, depth ~10.82 m.

Severity: CRITICAL | Probability: 100.0%

Peak Discharge M3S: 1657.000
Wave Celerity Ms: 34.500
Max Depth M: 10.820
Channel Width M: 20.000

DOWNSTREAM ARRIVAL TIMES

Settlement / Region	Arrival Time (min)	Urgency
village_A	2.4	IMMEDIATE
town_B	7.2	IMMEDIATE
city_C	18.5	URGENT

RECOMMENDED ACTIONS

1. Deploy slope monitoring sensors on identified failure zones.
2. Restrict access to hillside roads and settlements.
3. Station rapid response teams at channel blockage point.
4. Monitor lake water level continuously — install early warning gauge.
5. Issue immediate downstream evacuation order.
6. Notify emergency services — peak discharge exceeds safe channel capacity.
7. Activate flood wave sirens in downstream settlements.
8. Open emergency shelters on high ground above estimated wave depth.

This report was generated by the Nerolith FLOOD-AI Cascade Analysis Engine. Results are based on physics simulation outputs and should be validated against real-time field observations before operational decisions are made.